

GE261004 DESIGN THINKING

Unit III – Ideation and Creative Thinking · Presentation 2 of 6

Brainstorming and Brainwriting

Osborn's rules · what each rule is for · why brainstorming underperforms · brainwriting 6-3-5 · three other written methods

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In this presentation

- Where brainstorming came from
- The seven rules
- What the rules are for
- Why brainstorming often fails
- Brainwriting 6-3-5
- How a round works
- The two methods compared
- Three other written methods

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Brainstorming

The oldest formal ideation method, and the most often done badly

Where brainstorming came from

- Alex Osborn, an advertising executive, developed it from 1939 and published it in Applied Imagination in 1953
- **Four original rules:** go for quantity, withhold criticism, welcome wild ideas, combine and improve
- He summarised the fourth as “1 + 1 = 3” – ideas breed on each other
- The Stanford d.school later added three rules about how the room itself behaves

KEY POINT Nothing in the method is new or difficult. Almost every failure comes from skipping the rules.

The seven rules

1

Defer judgement

no criticism, not yet

2

Encourage wild ideas

the tame ones come free

3

Build on others

say “yes, and...”

4

Stay on topic

one question at a time

5

One conversation

nobody talks over

6

Be visual

draw it, do not describe it

7

Go for quantity

aim for a hundred

Read them aloud at the start. A rule nobody has agreed to is not a rule.

What the rules are actually for

- **Deferring judgement** protects the ideas that sound wrong but are not
- **Wild ideas** widen the field; the tame ones then arrive for free
- **Building on others** is what makes the group worth more than the individuals
- **Being visual** makes an idea specific enough for somebody else to build on

KEY POINT Each rule exists to stop one particular failure. Name the failure and the rule stops sounding like etiquette.

Why brainstorming often fails

- **Production blocking** – only one person can speak, so the other five are waiting instead of thinking
- **Evaluation apprehension** – people hold back whatever might make them look foolish
- **Anchoring** – the whole room circles the first idea that was offered
- Diehl and Stroebe's review of 22 studies found groups brainstorming together produce fewer ideas than the same people working separately

KEY POINT The problem is not the rules. It is the talking – and that is precisely what brainwriting removes.

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Brainwriting

Everybody generating at once, in silence

Brainwriting 6-3-5

one worksheet, six rows of three

1			
2			
3			
4			
5			
6			



6

participants

3

ideas each

5

minutes

6

rounds

108 ideas

in thirty minutes, in silence

Six participants, three ideas, five minutes – six times over.

How a round works

- Bernd Rohrbach published the method in 1968: six people, three ideas, five minutes
- Each person writes three ideas on a row of the sheet, then passes it to the left
- The next person reads what is there and adds three more – extending or reacting
- Six rounds, thirty minutes, 108 ideas, and not one word spoken

ACTIVITY Run one full round now. Three ideas, five minutes, then pass left. Complete silence until the timer stops.

The two methods compared

	Brainstorming	Brainwriting
Medium	Spoken	Written
Turn taking	One at a time	Everyone at once
Quiet people	Often lost	Equally represented
The record	Needs a scribe	Records itself
Energy	High and social	Low and focused

Most good sessions use both: brainwriting to generate the field, brainstorming to build on it.

Three other written methods

Brainwriting pool

Sheets go into the middle of the table. Take any one, add to it, put it back.

Crazy Eights

Fold a sheet into eight boxes. Eight sketches in eight minutes – one a minute.

Round robin

One idea each, in turn, round the table, until somebody passes twice.

KEY POINT All three do the same job: remove the queue, so everybody generates at the same time.

In class: 6-3-5 on your own question

- Take your team's How Might We question from Unit II and write it at the top
- Six rows of three boxes on one sheet, one sheet per person
- Five minutes a round, then pass to the left – read before you write
- After six rounds, count the ideas and mark the five most unexpected

ACTIVITY Thirty minutes, in silence. Every team should leave with roughly a hundred ideas and five marked as wild.

THINK ABOUT IT

Your team has 108 ideas and forty minutes left. What do you do next?

- How would you group them without losing the odd ones?
- Who decides what counts as one idea rather than two?
- Which five would you protect from being voted out?
- What would you regret throwing away?

Summary

- Osborn published brainstorming in 1953; four rules, later extended to seven
- Every rule prevents one specific way a session goes wrong
- Talking causes blocking, apprehension and anchoring
- Rohrbach's 6-3-5, 1968: six people, three ideas, five minutes
- Six rounds give 108 ideas in thirty silent minutes
- Pool, Crazy Eights and round robin remove the queue the same way

Next: Mind mapping

- What a mind map is, and where it came from
- How to build one, and the rules that matter
- Mind maps and concept maps compared
- Turning a finished map into themes